

# RP: DLens: Closed-Loop AI Agents for Parsimonious Scientific Machine Learning in Dark Matter Gravitational Lensing

Anonymous Authors<sup>1</sup>

<sup>1</sup>Affiliation Masked for Review

Paper ID: 4026 (Regular Papers Track)

1st Symposium on Parsimonious Scientific Machine Learning (ParSciML-FSS26)

2026 AAAI Fall Symposium Series (Arlington, VA, USA)

## Abstract

Deploying deep neural networks to analyze simulated astrophysical observations often introduces substantial computational overhead and severe model over-parameterization. Standard scientific machine learning workflows frequently rely on large computer vision backbones that overfit limited training sets and require labor-intensive hyperparameter tuning. We present **DLens**, a multi-agent framework designed according to Parsimonious Scientific Machine Learning (ParSciML) principles to identify low-mass dark matter substructures in strong gravitational lensing observations. DLens integrates physical lensing simulation backends (`lenstronomy` and `pyHalo`) with an interactive Large Language Model (LLM) architecture search and experiment planning loop. To prevent prompt context inflation, DLens uses structured Pydantic data schemas to decouple high-level experimental reasoning from raw image arrays. Under sample-constrained conditions (1,800 training images), DLens combines a tree-based architecture search across valid neural network candidate families with a ReAct optimization planner that inspects scalar validation metrics to apply targeted hyperparameter adjustments. Evaluated on a 3-class dark matter substructure benchmark (no substructure, cold dark matter, and ultralight axion dark matter), DLens discovers a compact convolutional neural network (`cnn_medium_s4`, 2.54M parameters) achieving 87.06% accuracy on a held-out test dataset. This improves upon a ResNet-34 baseline (81.00%, 21.28M parameters) by 6.1 percentage points while achieving an 8.4 $\times$  reduction in parameter scale (88.1% parameter reduction). The complete search and tuning pipeline runs in 21.0 minutes on a single workstation, demonstrating computational parsimony and reproducible workflow verification.

## 1 Introduction

Strong gravitational lensing provides a direct observational probe of dark matter distributions on sub-galactic scales (Alexander et al. 2020a,b; Toomey et al. 2021). Low-mass Cold Dark Matter (CDM) subhalos and ultralight axion dark matter interference patterns induce subtle localized perturbations in lensed arc morphologies. Identifying these astrometric distortions from smooth main galaxy light profiles requires high-resolution imaging and sensitive statistical classifiers.

Machine learning models trained on synthetic observations generated by physical simulators such as `lenstronomy` (Birrer and Amara 2018) and `pyHalo` (Gilman et al. 2020) have shown strong diagnostic potential (Cranmer, Brehmer, and Louppe 2020). However, practical scientific machine learning (SciML) workflows face three key operational challenges:

1. *Sample Inefficiency*: Standard deep neural networks typically require large synthetic training sets to converge, whereas real space-based observations remain sample-limited.
2. *Architectural Over-Parameterization*: Practitioners frequently adopt standard vision backbones (e.g., ResNet-34/50 with  $> 20\text{M}$  parameters) that tend to overfit small training datasets and increase inference latency.
3. *Manual Tuning Overhead*: Designing candidate networks and resolving training issues like overfitting often demand manual hyperparameter sweeps across multi-GPU compute resources.

Recent advances in Large Language Model (LLM) agents demonstrate their ability to automate code generation and experiment planning (Yao et al. 2023; Lu et al. 2024). Unstructured LLM code generation, however, can introduce syntax errors, invalid simulation parameters, or excessive prompt token usage when handling raw numerical arrays.

We address these challenges with **DLens**, an automated framework designed around Parsimonious Scientific Machine Learning (ParSciML) principles. DLens prioritizes sample efficiency, model compactness, computational parsimony, and verifiable execution. Rather than relying on large pre-trained backbones or unguided prompt loops, DLens enforces typed data contracts between physical simulation modules, neural architecture builders, evaluation metrics, and a closed-loop experiment planner.

The primary contributions of this work are:

- **Type-Safe Contract Workflows**: Structured Pydantic schemas (`SimConfig`, `PlannerDecision`) isolate language reasoning from numerical image data, keeping raw arrays out of language prompts while ensuring parameter integrity.
- **Sample-Efficient Tree Architecture Search**: A candidate generator, structural filter, and brief exploratory evaluation loop discover compact, task-matched architectures

under data-constrained conditions (1,800 training images).

- **Closed-Loop ReAct Optimization:** An experiment planner monitors scalar validation metrics (accuracy, ROC-AUC, confusion matrices) to execute targeted hyperparameter modifications that resolve generalization gaps.
- **Empirical Efficiency and Verifiability:** On a 3-class dark matter substructure benchmark, DLens discovers a compact network (`cnn_medium_s4`, 2.54M parameters) achieving 87.06% held-out test accuracy (+6.1% over a 21.28M parameter ResNet-34 baseline, representing an  $8.4\times$  parameter reduction) in 21.0 minutes of compute on a standard workstation.

## 2 Physical Foundations and Lensing Simulator

Gravitational lensing phenomena are governed by the lens equation mapping unlensed source coordinates  $\beta$  to observed image plane coordinates  $\theta$ :

$$\beta = \theta - \alpha(\theta) = \theta - \nabla_{\theta}\psi(\theta), \quad (1)$$

where  $\alpha(\theta)$  denotes the 2D deflection angle vector and  $\psi(\theta)$  represents the 2D projected gravitational lensing potential. The local convergence  $\kappa(\theta)$ , representing normalized surface mass density, satisfies Poisson's equation in two dimensions:

$$\kappa(\theta) = \frac{1}{2} \nabla_{\theta}^2 \psi(\theta) = \frac{\Sigma(\theta)}{\Sigma_{\text{crit}}}, \quad (2)$$

where  $\Sigma_{\text{crit}} = \frac{c^2}{4\pi G} \frac{D_s}{D_l D_{ls}}$  defines the critical surface mass density given angular diameter distances to lens  $D_l$ , to source  $D_s$ , and between lens and source  $D_{ls}$ .

The primary lensing potential is modeled using a Singular Isothermal Ellipsoid (SIE) profile representing the host galaxy. To test cosmological dark matter hypotheses, localized substructure perturbations  $\delta\kappa(\theta)$  are superimposed on the SIE background:

$$\kappa_{\text{total}}(\theta) = \kappa_{\text{SIE}}(\theta) + \delta\kappa(\theta). \quad (3)$$

DLens evaluates three distinct physical substructure classes:

1. **Smooth Main Lens** (`no_sub`): Smooth SIE profile with zero localized substructure perturbations ( $\delta\kappa = 0$ ).
2. **Cold Dark Matter Subhalos** (`cdm`): Discrete, localized dark matter subhalos following a Navarro-Frenk-White (NFW) mass density profile with a subhalo mass function:

$$\frac{dN}{dM_{\text{sub}}} \propto M_{\text{sub}}^{-\gamma}, \quad \gamma \approx 1.9, \quad (4)$$

sampled across the mass range  $M_{\text{sub}} \in [10^6, 10^{10}] M_{\odot}$  with total host halo mass  $M_{\text{halo}} = 10^{12} M_{\odot}$ .

3. **Ultra-Light Axion Dark Matter** (`axion`): Wave-like dark matter fields ( $m_a \sim 10^{-22}$  eV) producing localized quantum interference fringes and granular vortices. The density fluctuations  $\delta\kappa_{\text{axion}}$  are generated by field superposition:

$$\psi_{\text{axion}}(\mathbf{r}, t) = \sum_j a_j e^{i(\mathbf{k}_j \cdot \mathbf{r} - \omega_j t + \phi_j)}, \quad (5)$$

yielding characteristic interference granules with vortex masses  $M_{\text{vortex}} \approx 3 \times 10^{10} M_{\odot}$ .

Simulated lensed images are generated on a  $150 \times 150$  grid with a pixel scale of  $0.05''/\text{pixel}$ , incorporating Gaussian source light distributions, elliptical shear components, and Point-Spread Function (PSF) convolution matching space-based optical observations.

## 3 System Architecture and Parsimonious Contracts

DLens separates language reasoning from numerical array processing by employing specialized agents that communicate via structured Pydantic schemas.

### 3.1 Pydantic Schema Decoupling

Physical simulators generate single-channel floating-point image arrays ( $150 \times 150$  pixels). Streaming numerical arrays directly into language model prompts creates substantial token overhead and degrades spatial reasoning. DLens decouples reasoning from array manipulation by encapsulating simulation parameters inside typed Pydantic schemas:

$$\text{SimConfig} = \left\{ \begin{array}{l} \text{substructure\_type} \in \{\text{no\_sub}, \text{cdm}, \text{axion}\}, \\ M_{\text{halo}} = 10^{12} M_{\odot}, \quad z_{\text{lens}} = 0.5, \quad z_{\text{source}} = 1.5 \end{array} \right\} \quad (6)$$

The Data Simulation Agent executes physical simulation routines within isolated environments, saving image arrays directly to disk storage. The agent returns a lightweight meta-data object (`SimOutput`) containing local file references, image checksums, and scalar simulation parameters, ensuring that raw image arrays do not enter the language model context.

### 3.2 Tree Architecture Search for Model Parsimony

Instead of defaulting to large standard vision networks, DLens searches systematically for compact, task-matched architectures:

1. **Candidate Proposals:** An LLM candidate generator proposes  $N = 10$  candidate specifications as `ArchitectureSpec` objects (stage depths 2–4, channel widths 8–1024). Each specification undergoes automated structural validation prior to PyTorch code generation.
2. **Structural Evaluation:** An LLM evaluator ranks the proposals based on architectural suitability for  $150 \times 150$  lensing images, passing the top  $K = 4$  candidate specifications to empirical validation.
3. **Exploratory Validation and Pruning:** Selected candidates undergo brief exploratory training runs (10 epochs) on the training set. The architecture exhibiting the highest validation performance is chosen for full closed-loop optimization.

### 3.3 Closed-Loop ReAct Experiment Planning

Following initial model selection, the network enters an iterative optimization loop governed by a ReAct planner:

1. **Metric Observation:** The planner receives scalar evaluation summaries: training accuracy, validation accuracy, macro ROC-AUC, a  $3 \times 3$  confusion matrix, and per-class precision/recall values.
2. **Optimization Decisions:** The planner evaluates performance metrics to identify potential generalization issues (such as a gap between training and validation accuracy) and generates a `PlannerDecision` schema specifying a single hyperparameter adjustment (learning rate, weight decay, dropout rate, data augmentation, or early stopping patience).
3. **Automated Stop Criteria:** Optimization stops automatically when the generalization gap  $\text{Acc}_{\text{train}} - \text{Acc}_{\text{val}} \leq 0.06$  (given validation accuracy exceeds chance level), or if validation accuracy fails to improve by  $\Delta = 0.005$  over two consecutive iterations.

## 4 Experimental Setup

**Dataset Configuration:** We constructed a 9,000-image dark matter lensing dataset (3,000 images per class) following DeepLenseSim Model\_I specifications using `lenstronomy` 1.9.2 and `pyHalo` (Birrer and Amara 2018; Gilman et al. 2020). The dataset is divided into 7,200 training images and 1,800 validation images. To evaluate agent performance in data-constrained conditions, closed-loop optimization runs use an 1,800-image training subset (600 images/class). An independent test dataset of 1,800 fresh images generated with distinct random seeds (seeds 777/778/779, zero file overlap with training data) is evaluated once after tuning completes.

**Training and Computing Environment:** Neural models are implemented in PyTorch 2.13 and trained using Metal Performance Shaders (MPS) acceleration on an Apple M3 Max workstation. The LLM agents query `gpt-5.2` via the OpenAI API. System prompts explicitly restrict agent reasoning to quantitative scalar metric summaries.

## 5 Empirical Results

We evaluate two optimization runs under identical data splits and tuning rules:

1. **Searched Architecture:** The compact network discovered by the tree search (`cnn_medium_s4`) and tuned by the ReAct planner.
2. **ResNet-34 Baseline:** A standard ResNet-34 model (21.28M parameters) (He et al. 2016) tuned by the same ReAct planner.

### 5.1 Closed-Loop Overfitting Correction

Table 1 presents the optimization trajectories on the 1,800-image training subset. In iteration 0, both architectures exhibit overfitting: `cnn_medium_s4` reaches 99.94% training accuracy versus 82.33% validation accuracy (gap of 0.1761), while ResNet-34 reaches 98.39% training accuracy versus 75.17% validation accuracy (gap of 0.2322).

Table 1: Closed-loop optimization trajectories on the 1,800-image training subset. Both models overfit initially; the ReAct planner identifies the generalization gap from scalar metrics and enables data augmentation, closing the gap in iteration 1.

| Architecture                                  | Iteration | Train Acc | Val Acc       | Gap    | Val AUC       | Planner Decision         |
|-----------------------------------------------|-----------|-----------|---------------|--------|---------------|--------------------------|
| Searched Model ( <code>cnn_medium_s4</code> ) | 0         | 0.9994    | 0.8233        | 0.1761 | 0.9434        | Enable Data Augmentation |
|                                               | 1         | 0.8872    | <b>0.8600</b> | 0.0272 | <b>0.9717</b> | Stop (Gap Closed)        |
| ResNet-34 Baseline                            | 0         | 0.9839    | 0.7517        | 0.2322 | 0.8922        | Enable Data Augmentation |
|                                               | 1         | 0.8233    | <b>0.8167</b> | 0.0066 | 0.9283        | Stop (Gap Closed)        |

Table 2: Final evaluation on the 1,800-image held-out test dataset (evaluated once after tuning). The searched parsimonious model achieves higher accuracy and ROC-AUC while reducing parameter count by  $8.4\times$ .

| Architecture                                  | Parameters | Val Acc | Test Acc      | Val AUC | Test AUC      |
|-----------------------------------------------|------------|---------|---------------|---------|---------------|
| Searched Model ( <code>cnn_medium_s4</code> ) | 2.54M      | 0.8600  | <b>0.8706</b> | 0.9717  | <b>0.9700</b> |
| ResNet-34 Baseline                            | 21.28M     | 0.8167  | 0.8100        | 0.9283  | 0.9293        |

Based on the scalar metrics summary, the planner identifies the gap between training and validation accuracy and selects data augmentation (random flips and 90-degree rotations) to improve generalization. In iteration 1, validation accuracy increases by +5.0% for ResNet-34 (75.17%  $\rightarrow$  81.67%) and +3.7% for `cnn_medium_s4` (82.33%  $\rightarrow$  86.00%). The train-validation gap drops below 0.03, satisfying the automated stop criteria.

### 5.2 Held-Out Test Set Performance

Table 2 outlines final evaluation results on the 1,800-image held-out test dataset.

Empirical results demonstrate key performance characteristics:

- **Accuracy and Generalization:** The discovered `cnn_medium_s4` architecture reaches 87.06% test accuracy and a 0.9700 test ROC-AUC, outperforming the ResNet-34 baseline (81.00% test accuracy, 0.9293 ROC-AUC) by 6.1 percentage points.
- **Parameter Efficiency:** The searched network contains 2.54M parameters compared to ResNet-34’s 21.28M parameters, representing an  $8.4\times$  parameter reduction (88.1% parameter drop).
- **Pareto Optimization:** The search achieves a 2D Pareto hypervolume of 0.8678 across accuracy and parameter footprint trade-offs.
- **Computational Parsimony:** The complete optimization pipeline finishes in 21.0 minutes (1,261.2 seconds) on a single GPU workstation: 7.4 minutes for tree architecture search, 7.0 minutes per tuning run, and 6.1 seconds for test evaluation.

### 5.3 Per-Class Performance Breakdown

Table 3 presents the explicit  $3 \times 3$  confusion matrix for `cnn_medium_s4` evaluated on the 1,800-image test dataset (600 images per class). Summing the correctly predicted diagonal entries (600 for `no_sub`, 432 for `cdm`, and

Table 3: Confusion matrix for the searched model (cnn\_medium\_s4) on the 1,800-image held-out test dataset (600 images per class).

| Actual Class     | Predicted Class |            |            | Recall                |
|------------------|-----------------|------------|------------|-----------------------|
|                  | no_sub          | cdm        | axion      |                       |
| no_sub           | <b>600</b>      | 0          | 0          | 1.000                 |
| cdm              | 62              | <b>432</b> | 106        | 0.720                 |
| axion            | 9               | 56         | <b>535</b> | 0.892                 |
| <b>Precision</b> | 0.894           | 0.885      | 0.835      | <b>87.06% Overall</b> |

535 for axion) yields 1,567 correct classifications out of 1,800 total images, resulting in the overall test accuracy:

$$\text{Test Accuracy} = \frac{600 + 432 + 535}{1800} = \frac{1567}{1800} \approx 87.06\%. \quad (7)$$

The per-class metrics reflect high diagnostic specificity:

- no\_sub: Precision 0.894, Recall 1.000 (row [600, 0, 0], zero false negatives for smooth profiles).
- cdm: Precision 0.885, Recall 0.720 (row [62, 432, 106]).
- axion: Precision 0.835, Recall 0.892 (row [9, 56, 535]).

#### 5.4 Workflow Verification and Reproducibility

To ensure auditable execution integrity:

- **Parameter Verification:** Structural static analysis over generated simulation scripts confirms complete alignment with specified SimConfig parameters and numerical output validity.
- **Data Lineage Tracking:** Cryptographic SHA-256 hashing verifies data lineage and tamper detection (0.0185 s verification latency) across datasets and checkpoints.
- **Domain Bounds Checking:** Parameter validation guards intercept out-of-bounds simulation requests (such as unphysical mass values), projecting parameters onto valid physical ranges in 0.0003 s.

## 6 Discussion

**Model Capacity in Data-Constrained Regimes:** In data-limited scientific applications (1,800 training images), large vision backbones like ResNet-34 tend to overfit rapidly due to high model capacity. The DLens architecture search identifies compact, shallower convolutional networks that match the information content of lensed features, mitigating overfitting while lowering memory requirements.

**Evaluator Selection Dynamics:** During candidate model ranking, prompt instructions encouraging low parameter counts occasionally led the LLM evaluator to favor overly constrained architectures (< 0.5M parameters) that underfit lensing patterns (Acc  $\approx$  0.64). Combining structural LLM ranking with empirical 10-epoch trial runs proved essential for selecting viable parsimonious architectures.

**Limitations:** Current limitations include: (1) single-seed dataset generation; (2) LLM API sampling variation; and (3) testing on simulated images without complex observational detector noise.

## 7 Ethical Statement

This research uses open astrophysical simulation tools (lenstronomy and pyHalo) and scientific machine learning libraries to advance sample-efficient and compute-efficient discovery. The methods and data presented do not involve human subjects, sensitive personal data, or dual-use security risks.

## 8 Conclusion

We presented **DLens**, a multi-agent framework for dark matter substructure classification in strong gravitational lensing built on Parsimonious Scientific Machine Learning principles. By pairing structured Pydantic data contracts with tree-based architecture search and closed-loop ReAct tuning, DLens identifies compact neural networks (2.54M parameters) achieving 87.06% test accuracy while providing an  $8.4\times$  parameter reduction over a standard ResNet-34 baseline in 21.0 minutes on a single workstation. Future work will extend DLens to multi-modal telescope observations and direct simulator parameter inference.

## References

- Alexander, S.; Gleyzer, S.; McDonough, E.; Toomey, M. W.; and Usai, E. 2020a. Deep Learning the Morphology of Dark Matter Substructure. *The Astrophysical Journal (ApJ)*, 893(1): 15.
- Alexander, S.; Gleyzer, S.; Parul, H.; Reddy, P.; Toomey, M. W.; Usai, E.; and Von Klar, R. 2020b. Decoding Dark Matter Substructure without Supervision. *arXiv preprint arXiv:2008.12731*.
- Birrer, S.; and Amara, A. 2018. lenstronomy: Multi-purpose gravitational lens modelling software package. *Physics of the Dark Universe*, 22: 189–201.
- Cranmer, K.; Brehmer, J.; and Louppe, G. 2020. The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences (PNAS)*, 117(48): 30055–30062.
- Gilman, D.; Birrer, S.; Nierenberg, A.; Treu, T.; Du, X.; and Benson, A. 2020. Warm dark matter chills out: constraints on the halo mass function with eight quadruple-image strong gravitational lenses. *Monthly Notices of the Royal Astronomical Society (MNRAS)*, 491(4): 6077–6101.
- He, K.; Zhang, X.; Ren, S.; and Sun, J. 2016. Deep Residual Learning for Image Recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778.
- Lu, C.; Lu, C.; Lange, R. T.; Foerster, J.; Clune, J.; and Ha, D. 2024. The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery. *arXiv preprint arXiv:2408.06292*.
- Toomey, M. W.; Alexander, S.; Gleyzer, S.; Parul, H.; and Usai, E. 2021. Deep Learning the Astrometric Signature of Dark Matter Substructure. *arXiv preprint arXiv:2112.12121*.
- Yao, S.; Zhao, J.; Yu, D.; Du, N.; Shafran, I.; Narasimhan, K.; and Cao, Y. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations (ICLR)*.